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Project Management

An introduction to Scrum and Agile

living.knowledge

Marius Martin

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Project Management – What is a Project?

"A project is a temporary endeavor undertaken to create a unique product, service, or result."
(Project Management Institute, 2008, p. 5)

→ **Temporary: Definite beginning and ending**

→ **Unique: Has not been done before**

→ **Product, Service, or result: Goal to be achieved**

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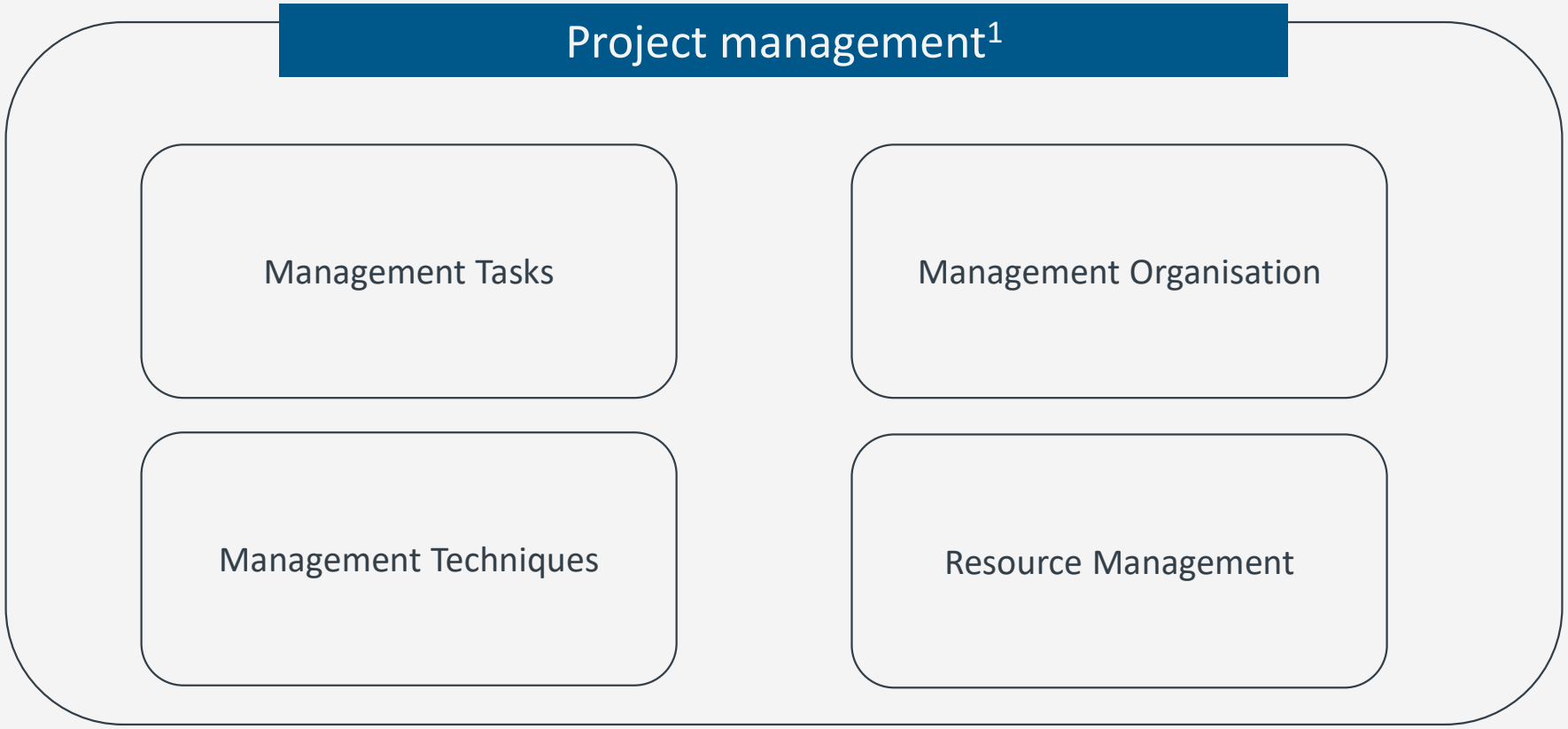
Project Management – What is Project Management?



[1] according to DIN 69901 (Helmold, 2023, p. 205)

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Project Management – Linear Methods

- Project is executed in a linearly ➔ No iterations
- Examples:
  - **(Extended) Waterfall Model**

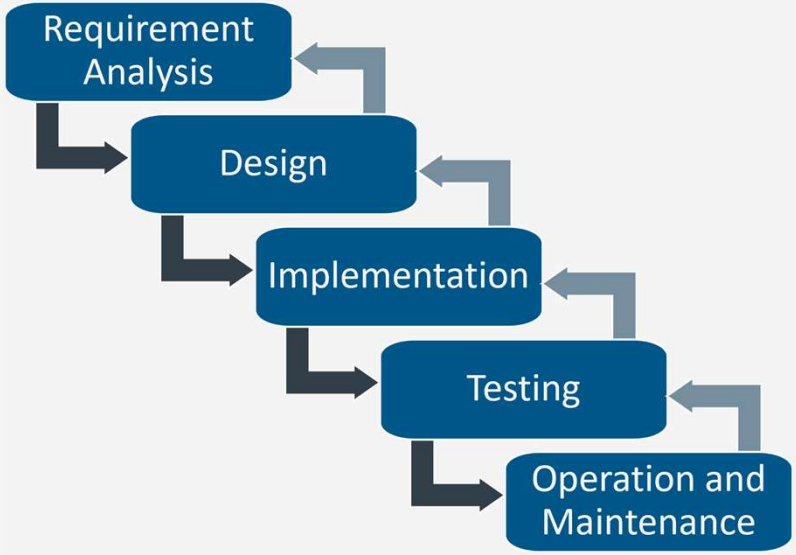

Figure 1: (Extended) Waterfall Modell (Adapted and extended from Adenowo & Adenowo, 2013, p. 429)

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Project Management – Linear Methods

- Project is executed in a linearly ➔ No iterations
- Examples:
  - **(Extended) Waterfall Model**
  - **V-Model**

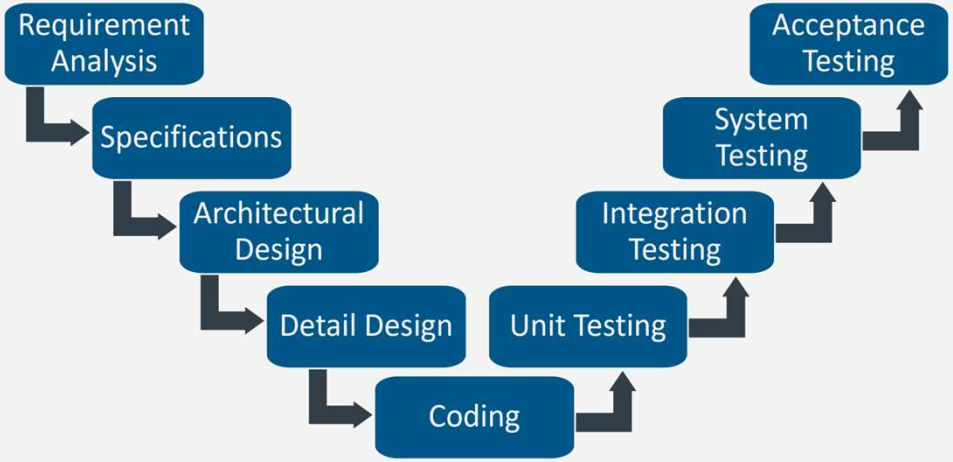

Figure 2: V-Model (Adapted from Mathur & Malik, 2010, p. 31)

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Project Management – Problems with Linear Methods

- Fixed Requirements
- Sequential Progression
- Client Value and Feedback

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Project Management – Agile Methods

- Iterative
- Delivery oriented
- Continuous feedback
- Examples:
  - **Kanban**

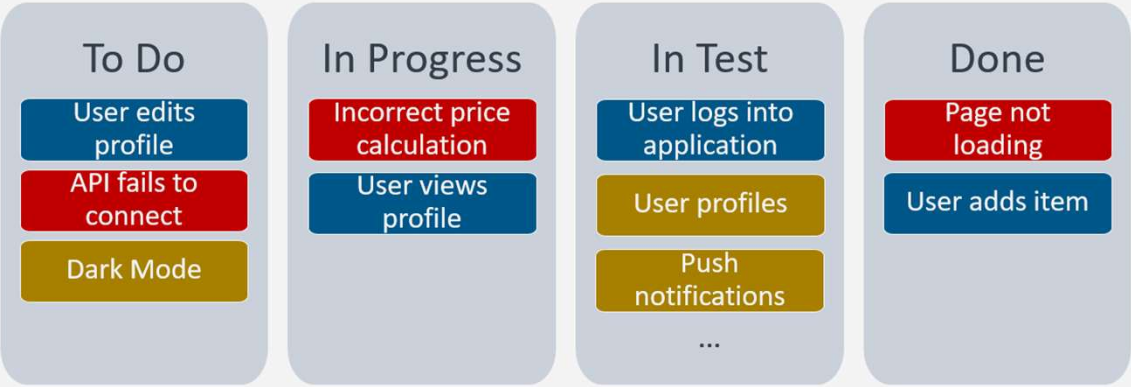

Figure 3: Example of a Kanban board

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Project Management – Agile Methods

- Iterative
- Delivery oriented
- Continuous feedback
- Examples:
  - **Kanban**
  - **Spiral Model**
  - **Scrum**

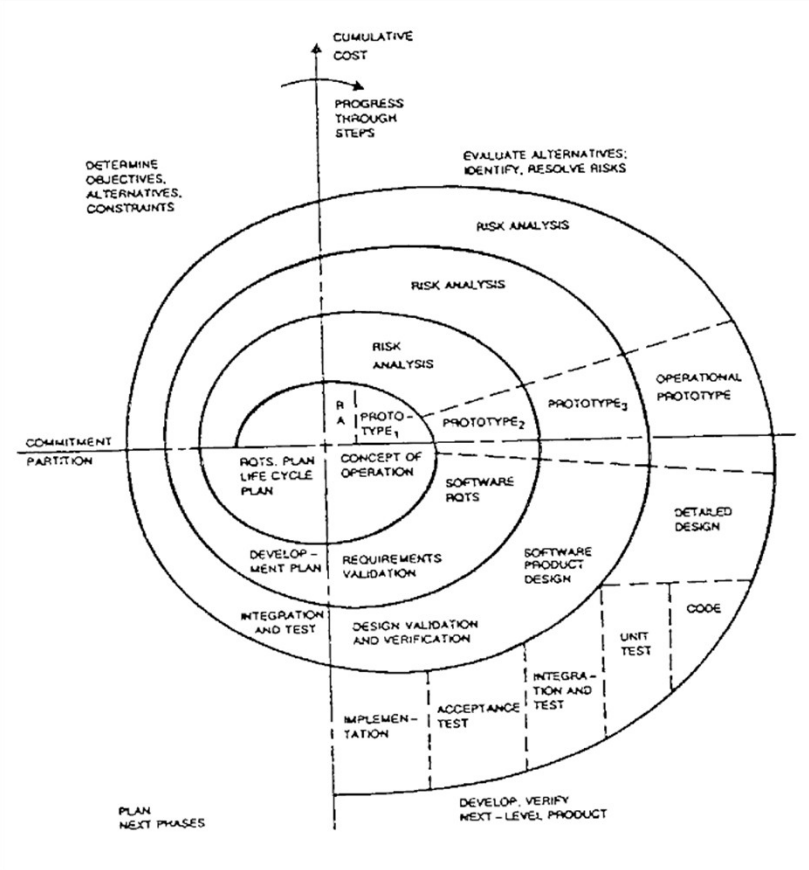

Figure 4: Spiral Model (Boehm, 1986, p. 25)

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Scrum Theory and Practise – What is Scrum?

"Scrum is a lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems" (Schwaber & Sutherland, 2020, p. 3)

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Scrum Theory and Practise – Paradigms

Empiricism

- Knowledge from experience
- Decisions from observations

Lean Thinking

- Focus on essentials
- Reduce waste

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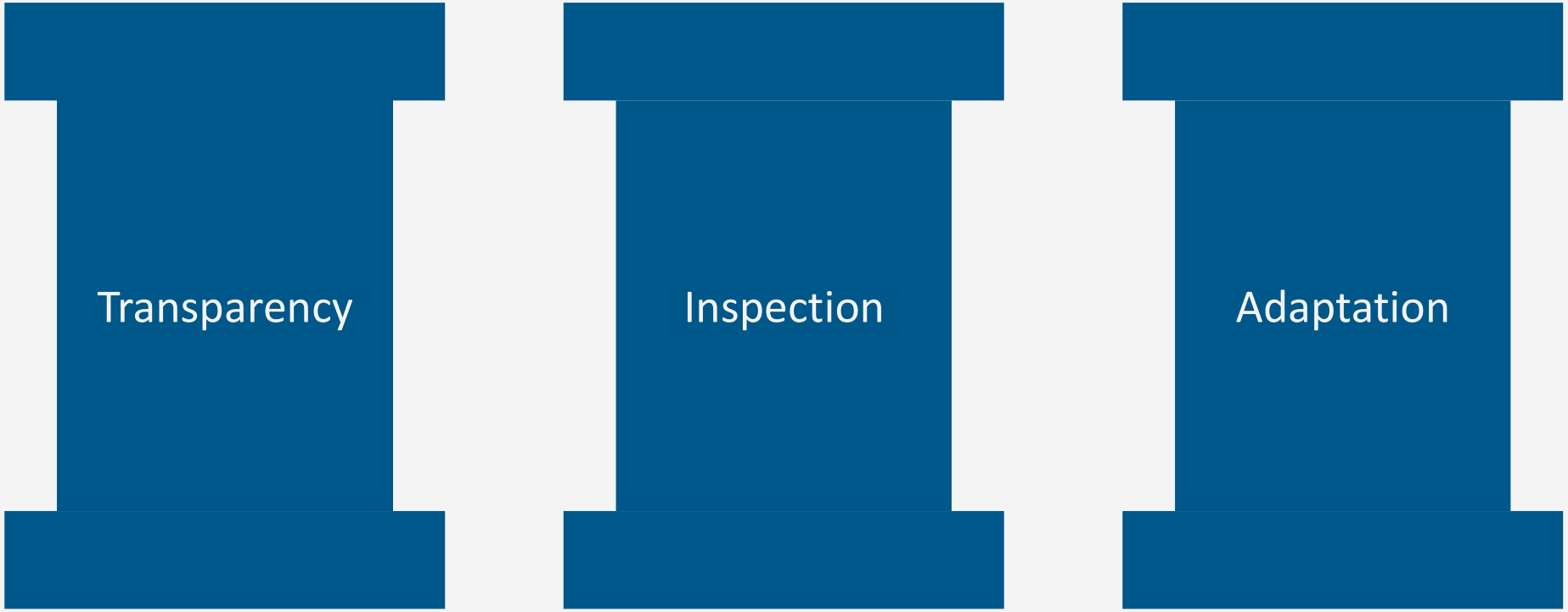
Scrum Theory and Practise – Pillars



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Scrum Theory and Practise – Values

- Commitment
- Focus
- Openness
- Respect
- Courage

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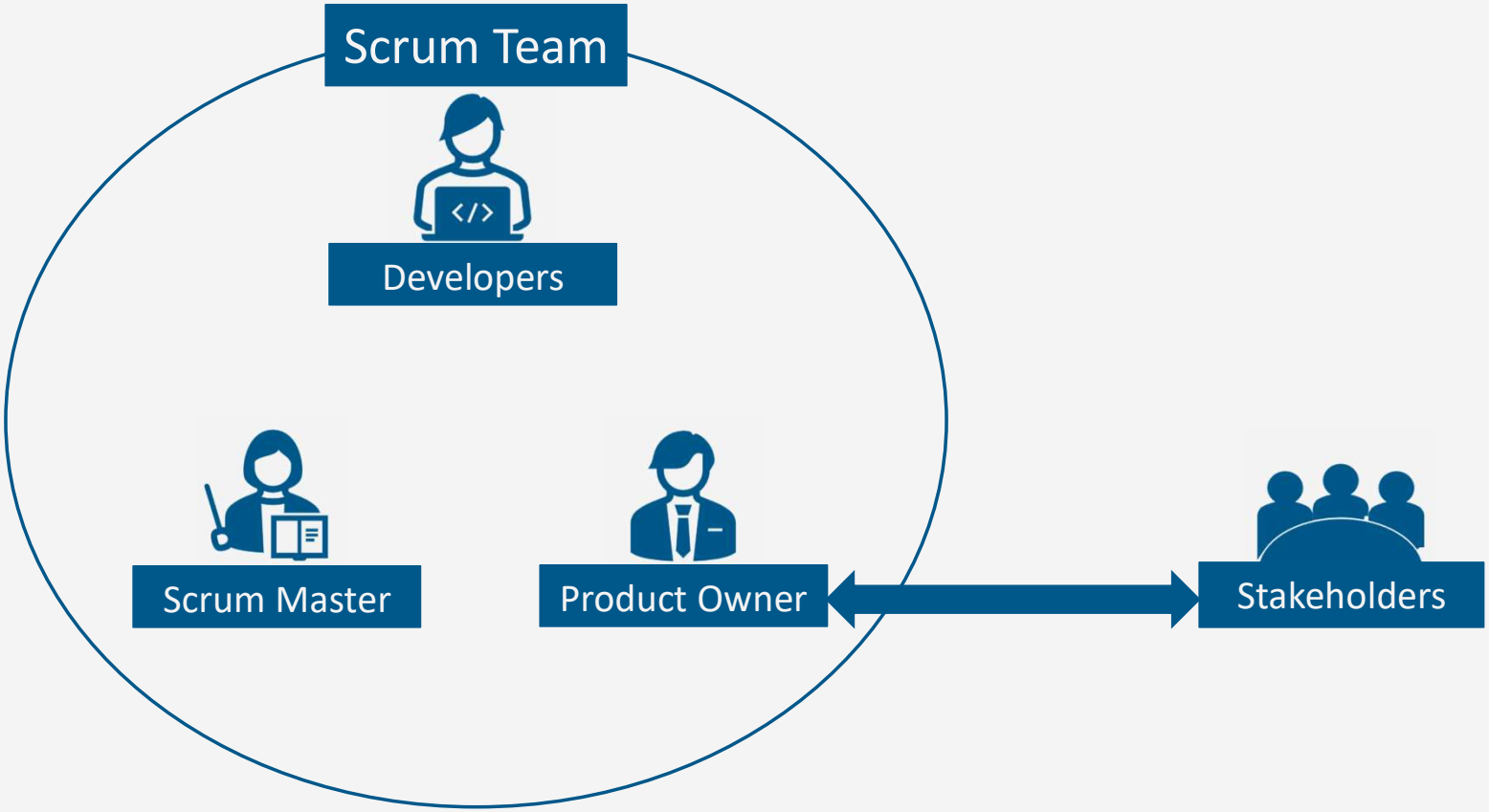
Scrum Theory and Practise – Team



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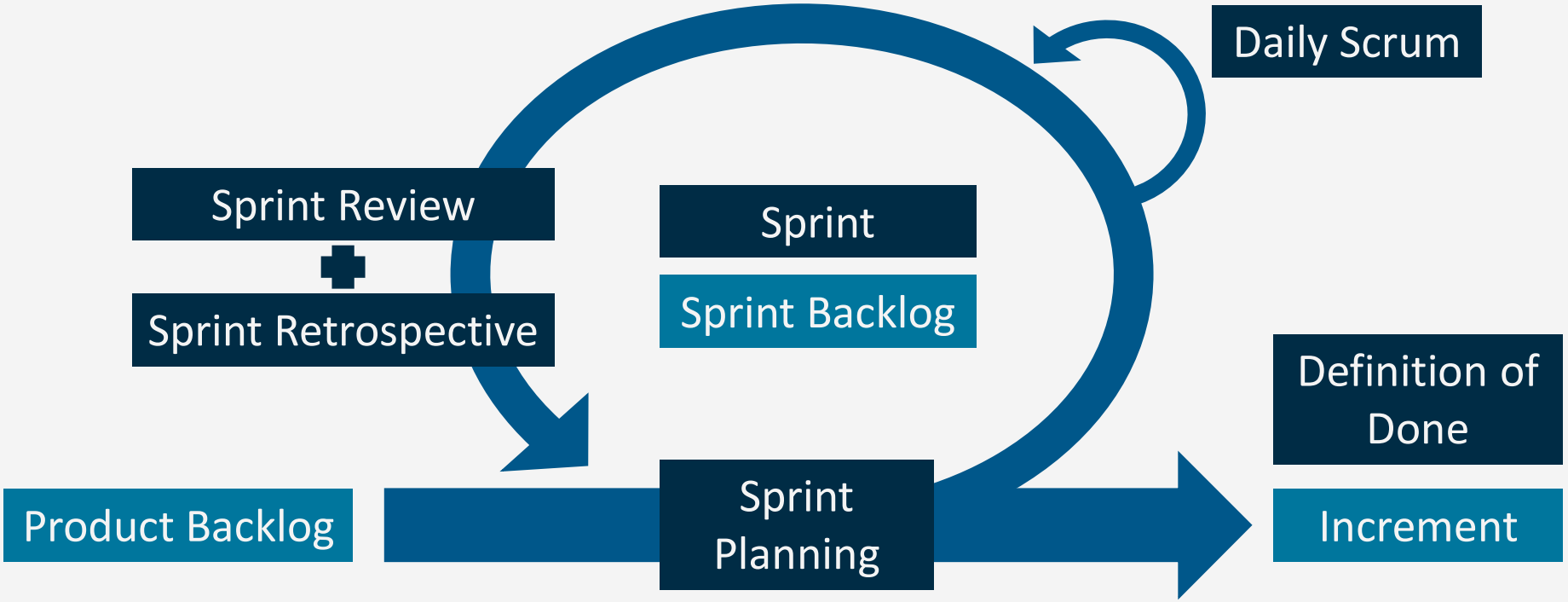
Scrum Theory and Practise – Artefacts and Events



Scrum Theory and Practise – Sprint Board

User Story

Feature

Bug

...

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Scrum Theory and Practise – Sprint Board

To Do	In Progress	In Test	Done
User logs into application			
API fails to connect			
Dark Mode			
…			

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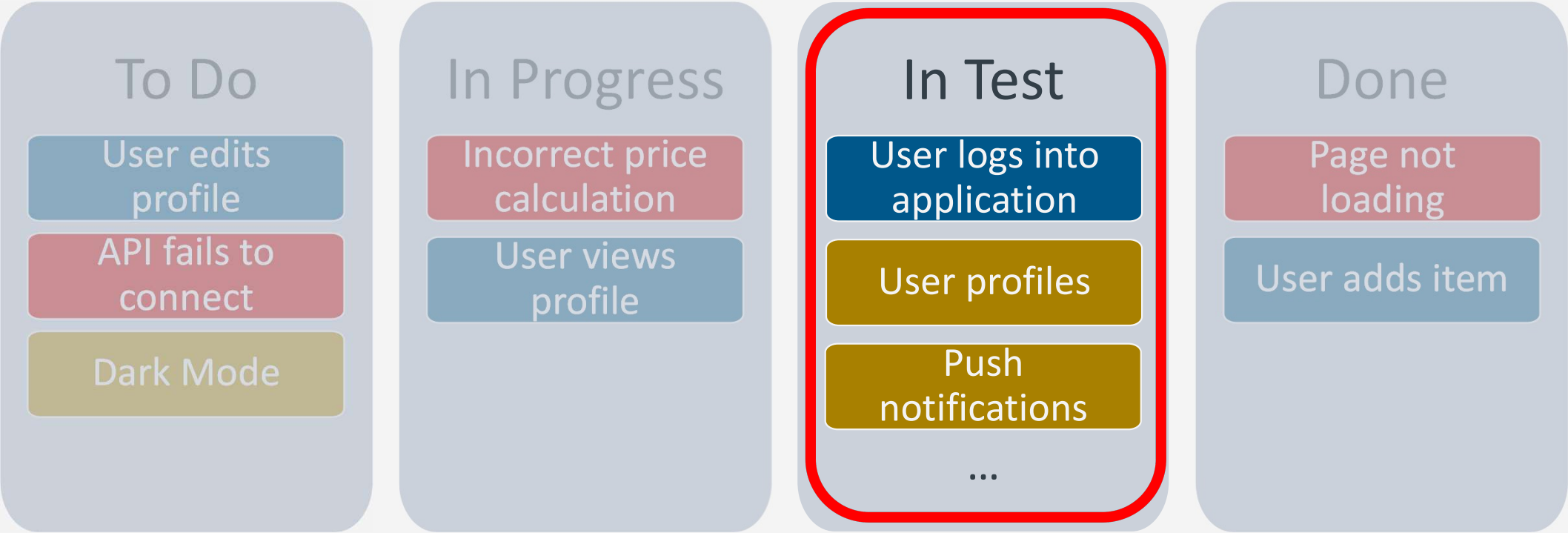
Scrum Theory and Practise – Sprint Board



Tools for Agile Software Development

- GitHub Issues
- Jira
- Trello
- Asana
- Azure DevOps

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Tools for Agile Software Development – Selection for today

- GitHub Issues
- Jira
- Trello

Tools for Agile Software Development – GitHub Issues

- Built-in issue tracking system in the repository
- Use Case: code-centric or open-source projects
- Costs: free

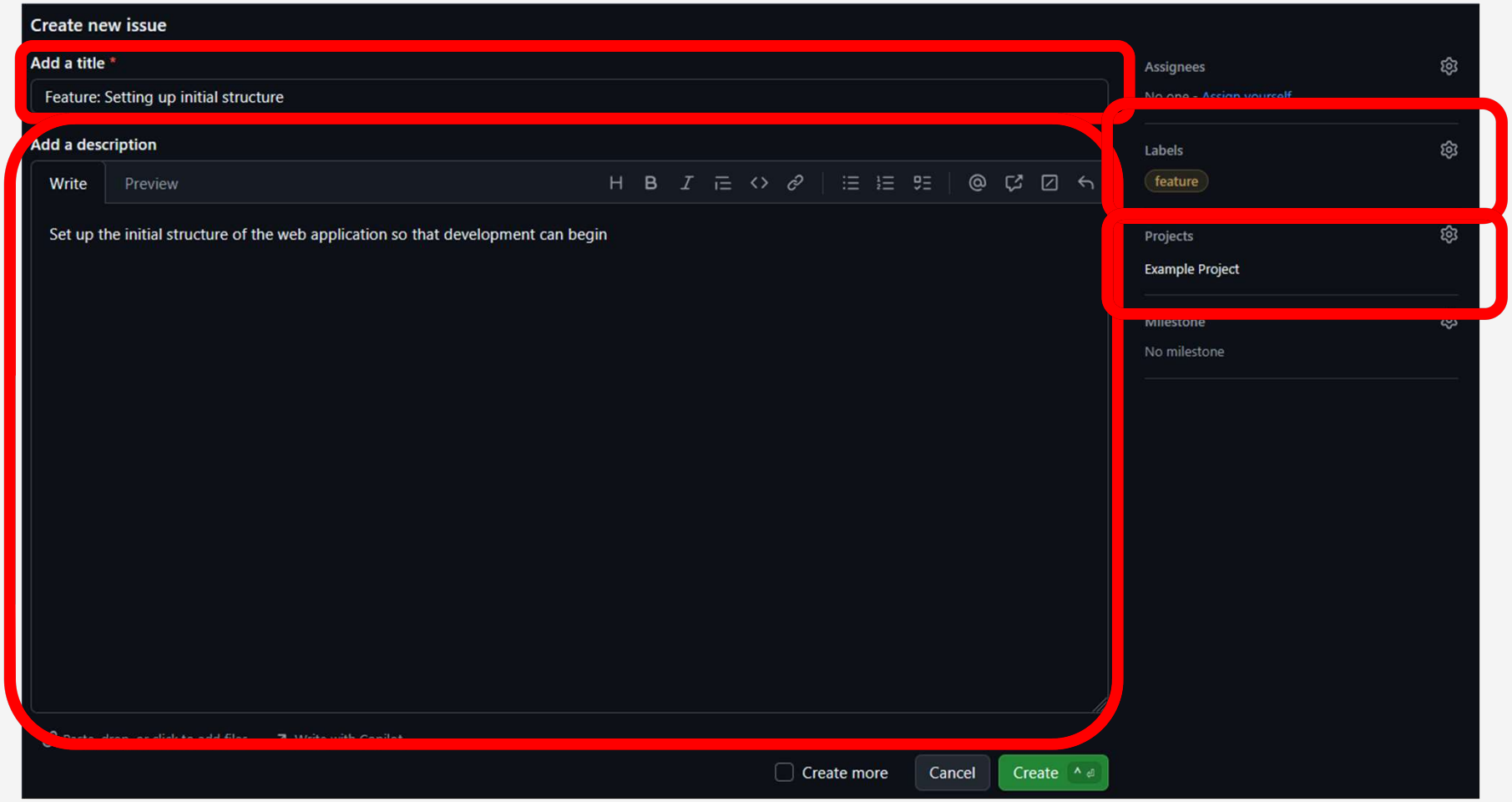

Figure 5: Create new issue in GitHub Issues

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Tools for Agile Software Development – GitHub Issues

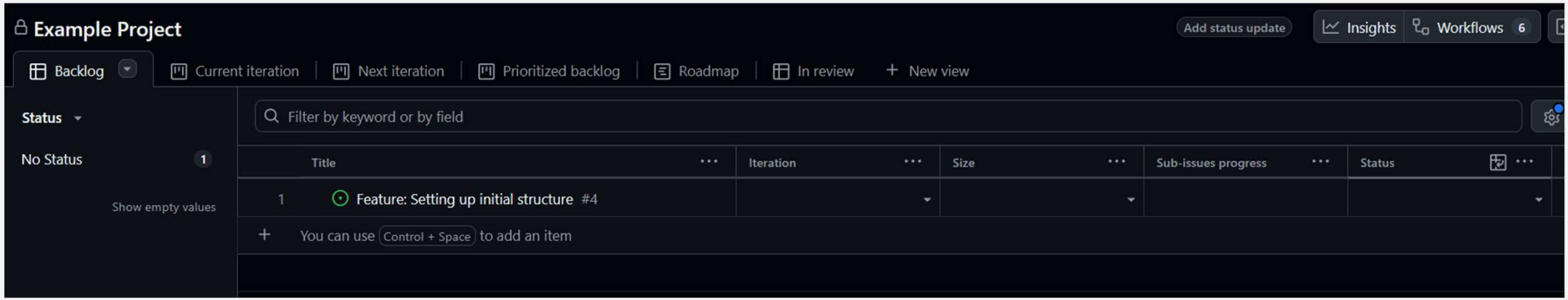

Figure 6: Backlog in GitHub Issues

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Tools for Agile Software Development – GitHub Issues

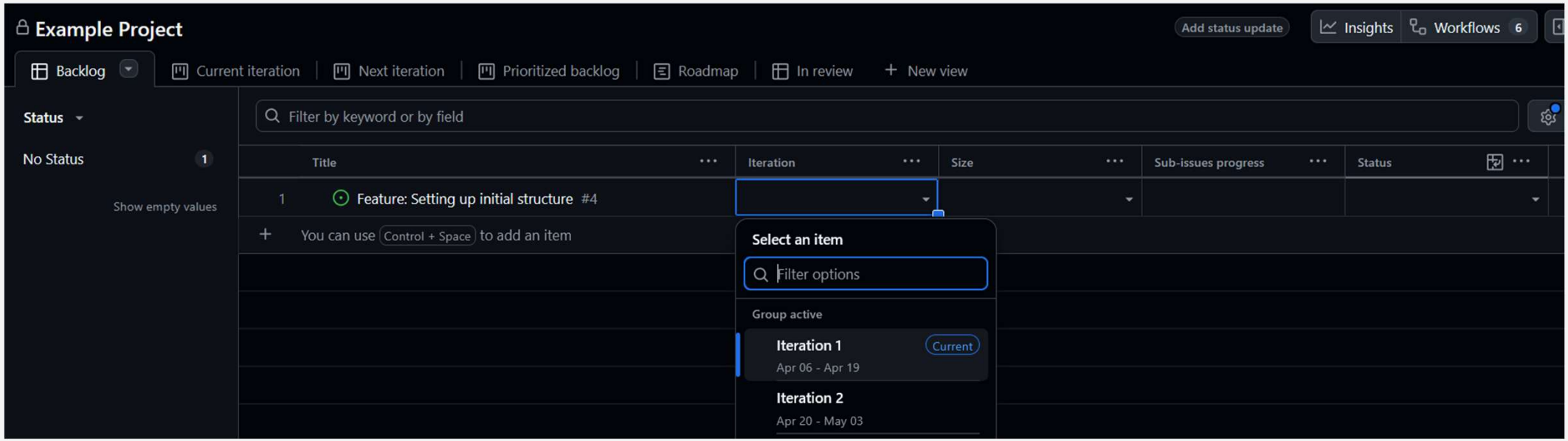

Figure 7: Add backlog-item to iteration in GitHub Issues

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Tools for Agile Software Development – GitHub Issues

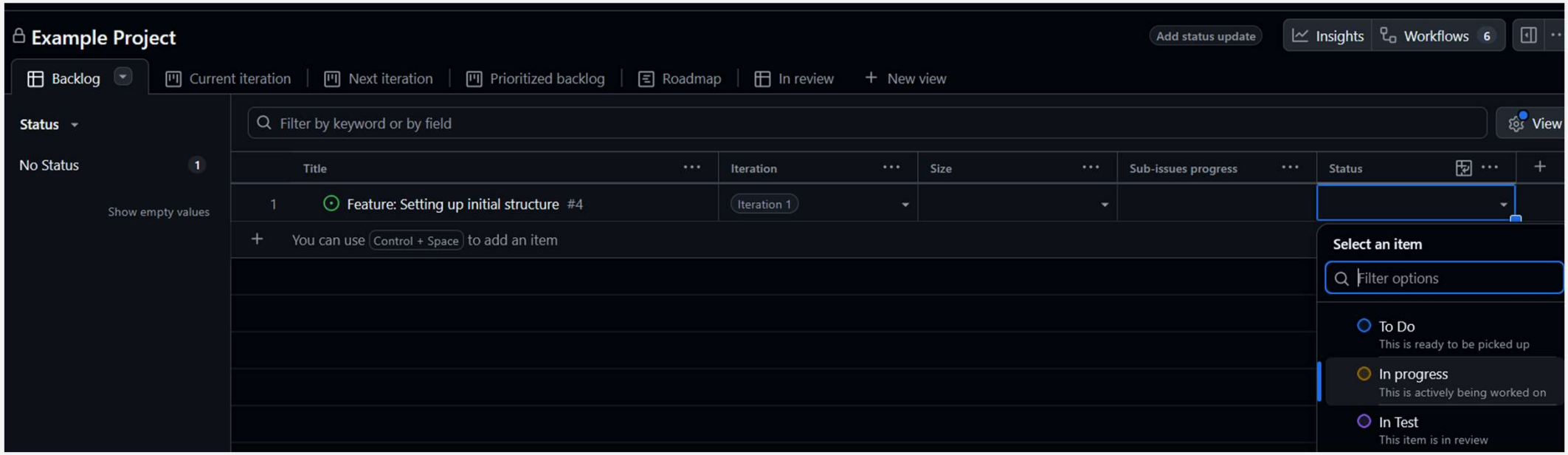

Figure 8: Editing status of backlog-item in GitHub Issues

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Tools for Agile Software Development – GitHub Issues

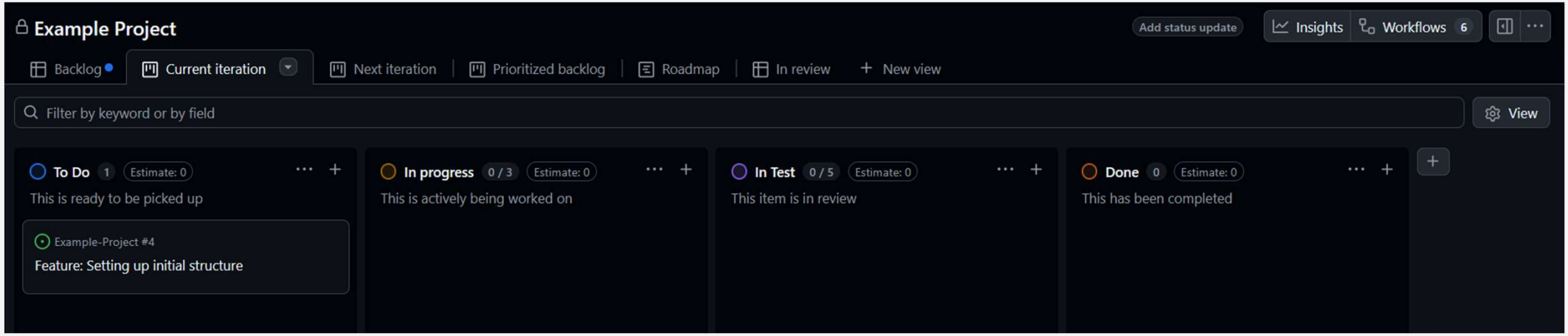

Figure 9: Sprint Board in GitHub Issues

Tools for Agile Software Development – GitHub Issues

- Advantages
  - **Integration in the repository**
  - **Simple setup and interface**
  - **Free**
- Disadvantages
  - **Limited reporting**
  - **Flat hierarchy**
  - **No workflow enforcement**

Tools for Agile Software Development – Jira

- Dedicated tool for project management
- Use-Case: Software projects with complex workflows
- Costs:
  - **free for 10 users**
  - **Standard, Premium and Enterprise for more users ($7,91 – $14.54 per user)**

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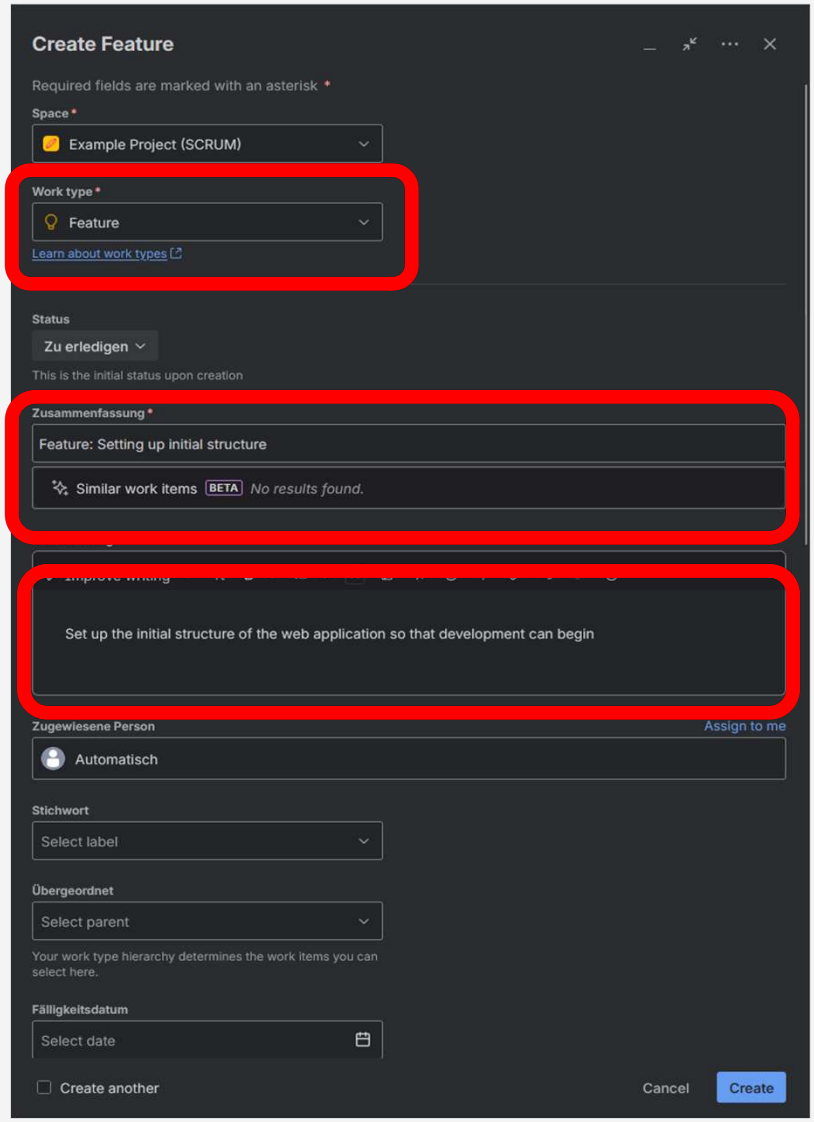

Figure 10: Create new Feature in Jira

Tools for Agile Software Development – Jira

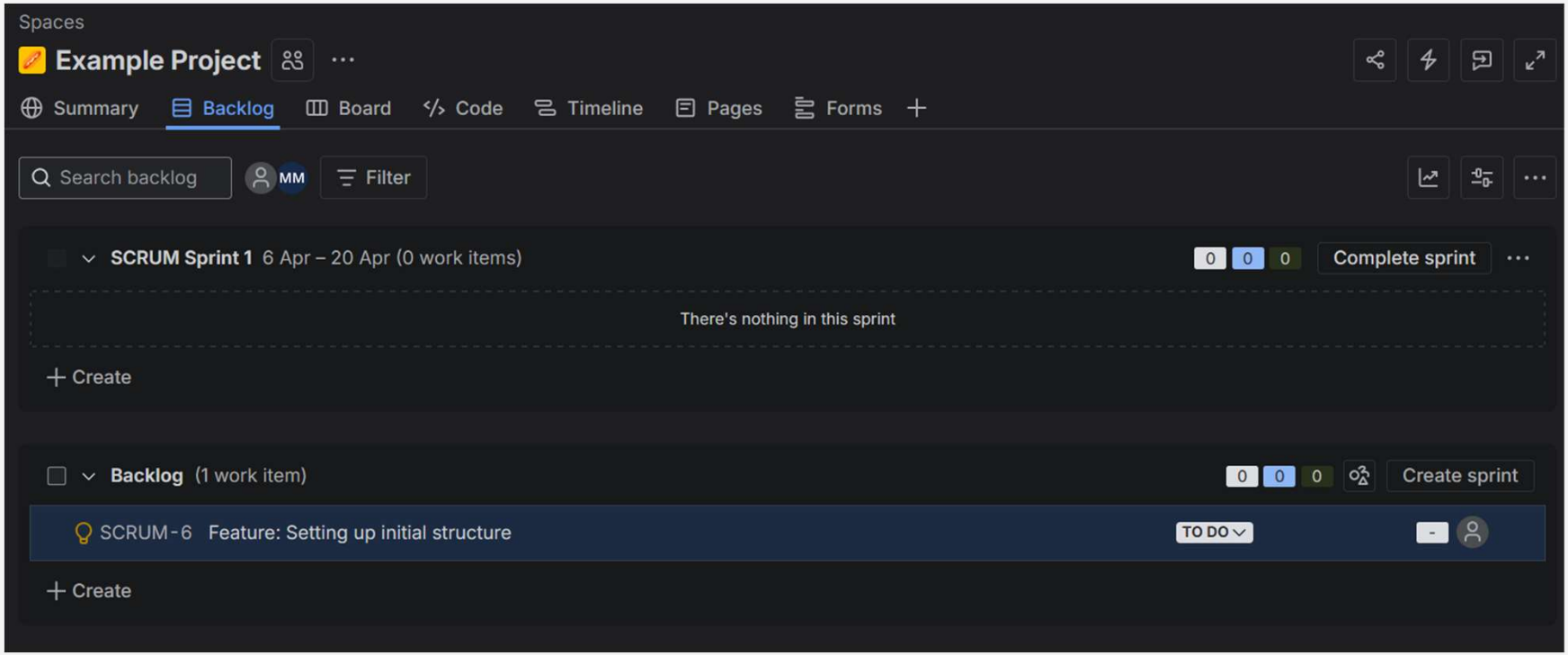

Figure 11: Backlog in Jira

Tools for Agile Software Development – Jira

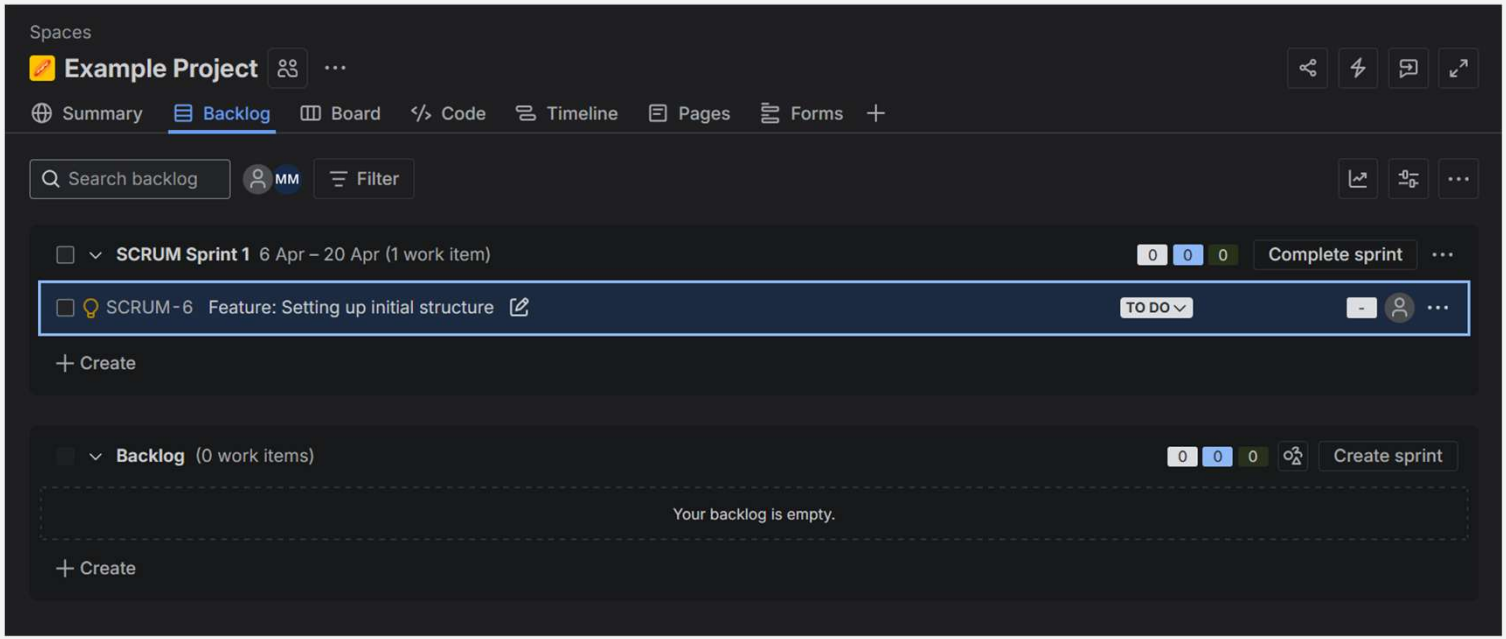

Figure 12: Add backlog-item to Sprint in Jira

Tools for Agile Software Development – Jira

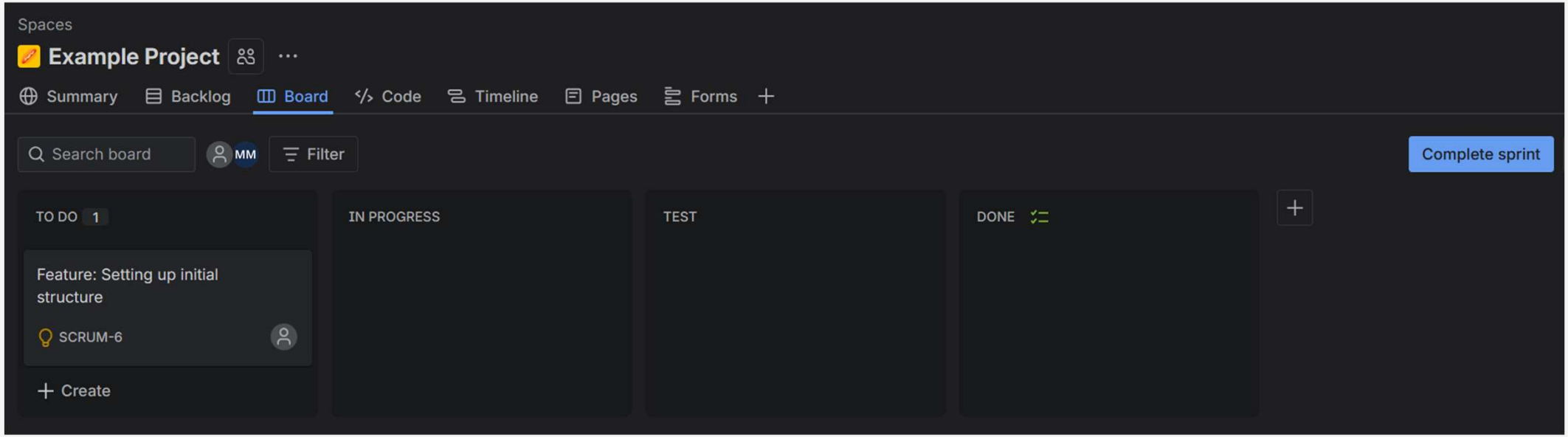

Figure 13: Sprint Board in Jira

Tools for Agile Software Development – Jira

- Advantages
  - **Customizable workflows**
  - **Advanced reporting capabilities**
  - **Large ecosystem**
- Disadvantages
  - **Complexity**
  - **Friction**
  - **Costs for larger teams**

Tools for Agile Software Development – Trello

- Dedicated tool for project management
- Use-Case: Software projects with simple workflows
- Costs:
  - **free for 10 users**
  - **Standard, Premium and Enterprise for more users ($5 – $17.50 per user)**

Tools for Agile Software Development – Trello

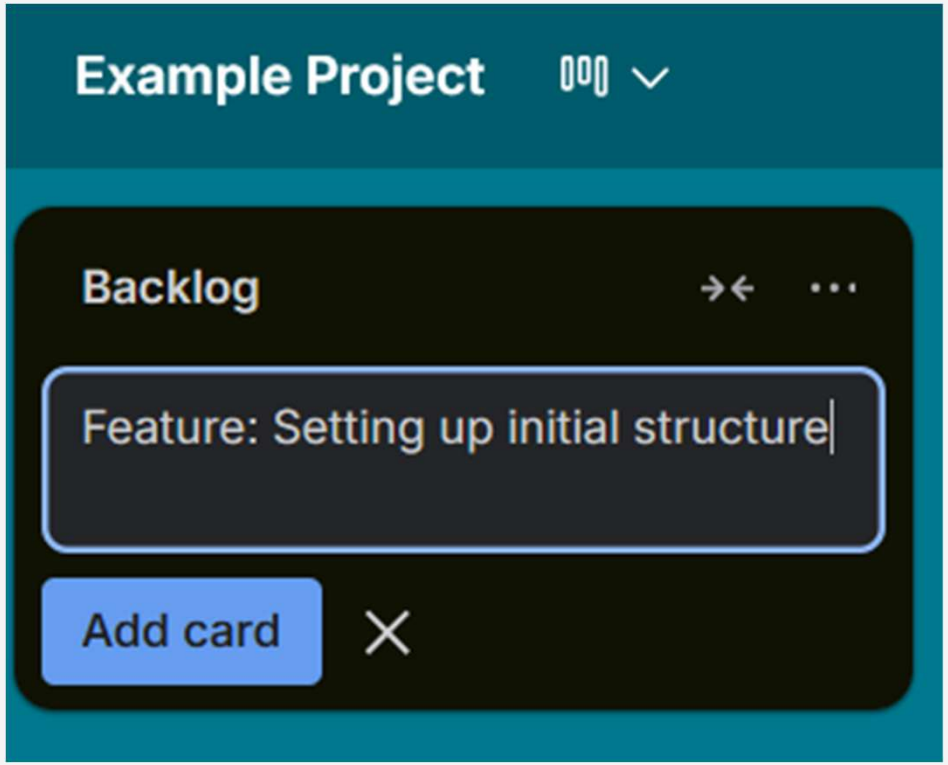

Figure 14: Create new card in Trello

Tools for Agile Software Development – Trello

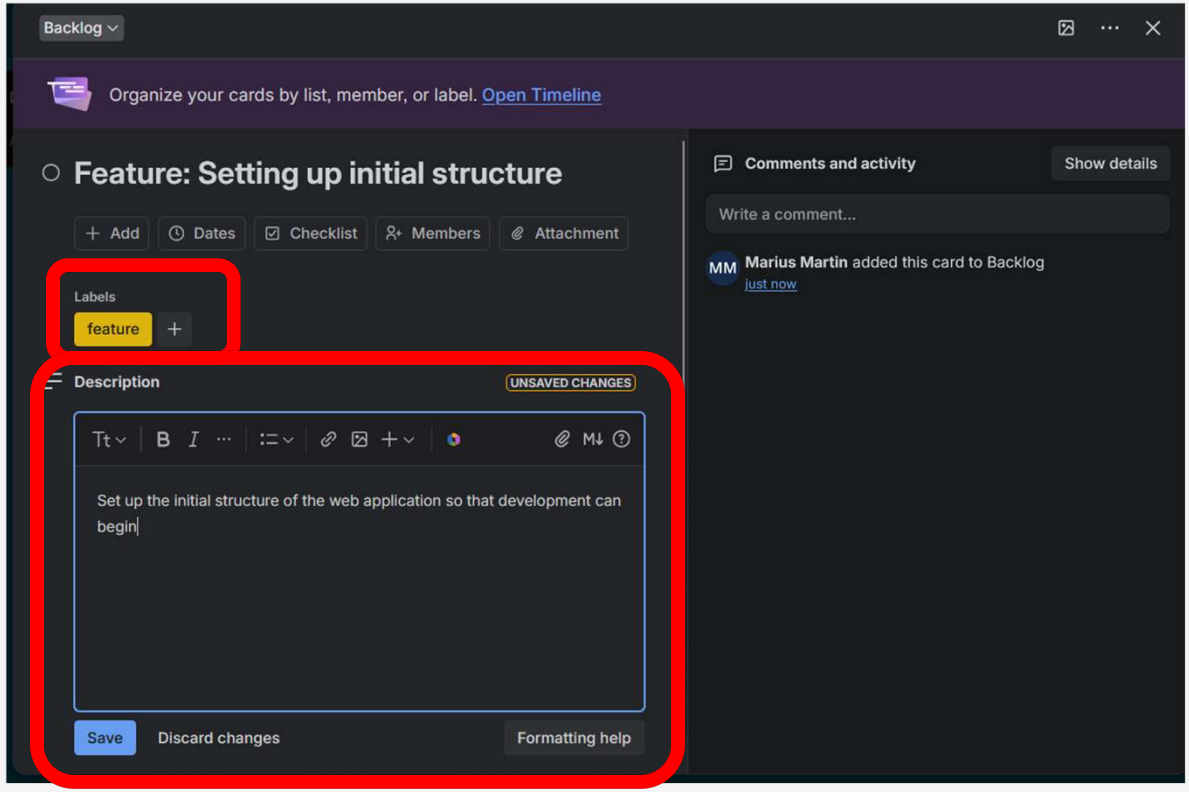

Figure 15: Add further information to card in Trello

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Tools for Agile Software Development – Trello

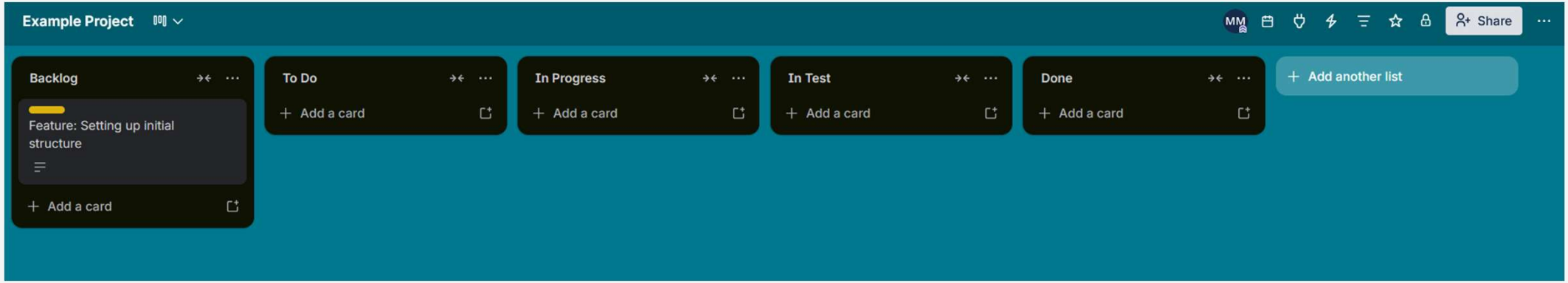

Figure 16: Backlog and Sprint Board in Trello

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Tools for Agile Software Development – Trello

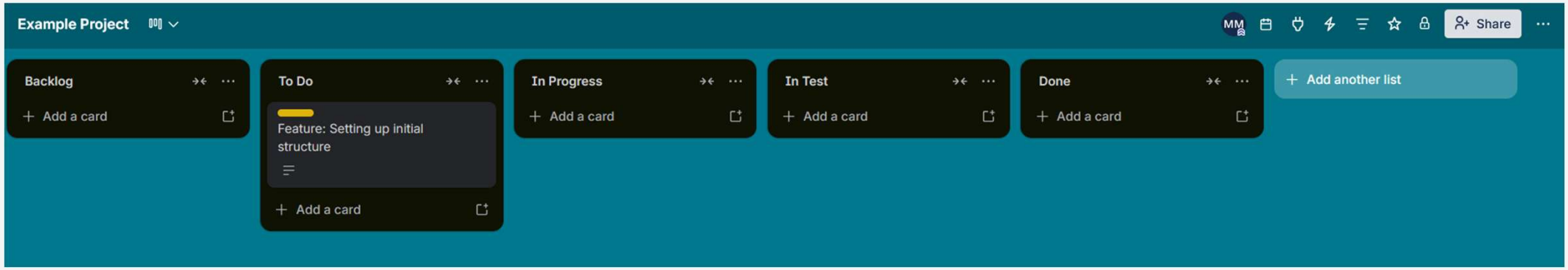

Figure 17: Add backlog-item to Sprint in Trello

Tools for Agile Software Development – Trello

- Advantages
  - **Intuitive board-based design**
  - **Customizable rules**
  - **Integration of other tools**
- Disadvantages
  - **Limited reporting**
  - **No real subtasks**
  - **Costs for larger teams**

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Thank you for your attention

Do you have any questions?

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Further Reading: Scrum Guide